Progress Report #2: Annual Panel with Hedonic Indices and Shift-Share Instrument

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Abstract

This report documents the second round of empirical analysis for the immigration–house prices project. Motivated by Round 1's null result (driven by a small 78-observation sample), we move to an annual city-year panel (461 observations) and construct five house price measures including hedonic indices. We build a new annual shift-share instrument from IPUMS 1900 settlement shares and digitized annual European immigration flows. The reduced-form estimates are **negative and significant**: the instrument coefficient on real log house prices is -2.71 (SE $= 0.95$, $p < 0.01$), consistent across simple and hedonic measures. However, controls attenuate the effect to zero, and the full 2SLS suffers from a weak first stage (KP $F = 1.03$). **Critical caveat:** post-hoc verification revealed a digitization error in the annual immigration data (Italy 1900–1909 understated by 20,000, or 1.05%), which affects instrument weights. These results require re-verification after data correction.

*University of Cologne. This project uses Claude Code (Anthropic) as an AI research assistant. See Section 7 for detailed attribution.

1 Motivation: From Round 1 to Round 2

Round 1 produced a null 2SLS result ($\hat{\beta} = -0.70$, SE $= 1.84$, $p = 0.70$) from a sample of only 78 observations (26 cities $\times$ 3 decades). The 95% confidence interval spanned $[-4.3, +2.9]$ — we could rule out neither large positive nor large negative effects. The key limitation was sample size: the Lyons house price data covers only $\sim$30 cities, and Tabellini's panel has only 3 decades.

Since we cannot increase the number of cities, Round 2 exploits the **annual frequency** of the Lyons data. Instead of averaging prices into three-year windows around census years, we use year-by-year price data from 1910 to 1930, yielding a potential 26 cities $\times$ 21 years $= 546$ observations. This requires constructing annual immigration measures to match.

2 New Data Construction

2.1 Annual National Immigration Flows

We digitized annual immigration counts by European country group from the *Historical Statistics of the United States, Colonial Times to 1970* (Bicentennial Edition), Series C 89–119: "Immigrants, by Country: 1820 to 1970." The source is a scanned PDF from the Census Bureau. The digitized data covers 11 European country groups (Great Britain, Ireland, Scandinavia, Other Northwestern, Germany, Poland, Other Central, USSR/Baltic, Other Eastern, Italy, Other Southern) for fiscal years 1900–1930.

Validation. We cross-checked decade totals against the DHS *Yearbook of Immigration Statistics* (2022 edition), Table 2. All three "all countries" decade totals match exactly: 1900–1909 = 8,202,388; 1910–1919 = 6,347,380; 1920–1929 = 4,295,510. Four of five country-decade cross-checks also match.

Digitization error (critical). Italy 1900–1909 is understated by 20,000 (CSV sum = 1,910,475 vs. DHS official = 1,930,475; 1.05% error). Additionally, the 1914 column sums exceed the reported Europe total by 1,000, suggesting one country value is slightly misread. The most likely single-year error is Italy 1903 (230,622 $\rightarrow$ $\sim$250,622). **This error directly affects the shift component of the instrument for the largest source country.**

2.2 Hedonic House Price Indices

We adapted Lyons et al.'s HPI construction code to produce city-level annual hedonic sale price indices for 1890–1935. Following their All-in-One (AIO) specification, we run separate per-city regressions:

$$\ln(\text{price})_{ict} = \sum_t \gamma_t \cdot \mathbf{1}[\text{year} = t] + \mathbf{X}_i'\boldsymbol{\delta} + u_{ict} \tag{1}$$

where $\mathbf{X}_i$ includes grouped total rooms, bedrooms, bathrooms, number of storeys, property type, machine-learning-predicted neighborhood, and lead paint indicator. The exponentiated year coefficients $\exp(\hat{\gamma}_t)$ form the index.

Results: 29 of 30 cities produced valid regressions (Las Vegas excluded for sparse pre-1935 data). R^2 ranges from 0.55 to 0.77, based on 2,000 to 31,000 transactions per city. We similarly constructed hedonic rent indices.

2.3 IPUMS Settlement Shares

Tabellini's replication data contains only the final instrument value, not the underlying 1900 settlement shares. We downloaded the IPUMS 1900 complete-count Census (∼76 million records, 2.1 GB compressed) via the IPUMS API and tabulated European-born residents by metropolitan area and birthplace country (mapped to the 11 Historical Statistics groups). The settlement share for country group k in city c is:

$$\text{share}_{ck} = \frac{n_{ck,1900}}{n_{k,1900}} \tag{2}$$

where $n_{ck,1900}$ is the number of group-k immigrants in city c and $n_{k,1900}$ is the national total.

3 Annual Shift-Share Instrument

The annual instrument is:

$$Z_{ct} = \frac{1}{\text{Pop}_{c,1910}} \sum_k \text{share}_{ck,1900} \times F_{kt} \tag{3}$$

where F_{kt} is annual immigration from country group k in year t (from the Historical Statistics) and $\text{Pop}_{c,1910}$ is the city population in 1910 (from Tabellini).

The annual instrument correlates 0.60 with Tabellini's decennial instrument at 1910, which is reasonable given different data sources (IPUMS vs. Tabellini's original aggregates)

and different temporal aggregation.

4 Results

4.1 Sample

The merged annual panel contains **461 city-year observations** (1910–1930) with non-missing instrument and sale prices. This is a 6-fold increase from Round 1's 78 observations.

4.2 Reduced Form (Main Result)

Table 1 reports reduced-form regressions of each price measure on the annual instrument, with city and year fixed effects. Standard errors are clustered at the metropolitan area level.

Table 1: Reduced Form: Effect of Annual Instrument on House Prices

	Simple Mean	Simple Median	Hedonic HPI	Hedonic RPI	Mean Rent
Panel A: City + Year FE					
Z_{ct}	-2.708^{***}	-2.707^{***}	-2.661^{**}	-1.732	-2.161
	(0.952)	(0.939)	(1.016)	(1.118)	(1.365)
N	461	461	461	462	462
R^2	0.870	0.851	0.740	0.779	0.726
Panel B: City + Year FE + Controls					
Z_{ct}	0.629	0.720	1.990	1.074	-0.637
	(1.039)	(1.078)	(1.198)	(1.903)	(2.282)
N	461	461	461	462	462
R^2	0.916	0.899	0.834	0.807	0.810

Standard errors clustered at metro area level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. All dependent variables in logs, deflated by CPI (1910 base). Controls: log 1900 city population × year, log 1900 immigration × year. Instrument based on IPUMS 1900 settlement shares × Historical Statistics annual flows (contains known digitization error for Italy 1900–1909).

In Panel A, three of five price measures show statistically significant negative effects at $p < 0.05$ or better. The point estimates are remarkably consistent: ~ -2.7 for sale prices. The hedonic HPI coefficient (-2.661, $p = 0.009$) is very close to the simple mean (-2.708, $p = 0.005$), suggesting that housing composition changes do not drive the result.

In Panel B, adding baseline controls attenuates all coefficients to statistical insignificance.

4.3 Full 2SLS

The IPUMS-derived endogenous variable (recent European arrivals from YRIMMIG / city population) yields a **weak first stage**: KP $F = 1.03$ without controls. The weakness arises because the endogenous variable is measured from Census stocks (survivors at Census time who report their arrival year), while the instrument captures annual national flows. Survivor bias, inter-city mobility, and YRIMMIG measurement error create a disconnect.

Given the weak first stage, 2SLS estimates are unreliable. **The reduced form in Panel A is the primary result** — it directly estimates the effect of predicted immigration (the instrument) on prices without requiring a valid first stage.

4.4 OLS

OLS coefficients on the immigrant share are positive for hedonic HPI (+0.099, $p = 0.01$) and rents (+0.134, $p < 0.001$), consistent with endogenous sorting: immigrants go where prices are rising. The reduced form reverses this sign, consistent with a causal negative effect masked by selection.

5 Data Quality Assessment

What passed verification:

- All "all countries" decade totals match DHS official figures exactly.
- 29 of 30 annual rows are internally consistent (country columns sum to Europe total).
- WWI pattern (sharp drop 1914–1918) and National Origins Act pattern (sharp drop after 1924, especially Southern/Eastern Europe) are historically correct.
- 4 of 5 country-decade cross-checks pass.

What failed:

- **Italy 1900–1909**: decade sum is 1,910,475 vs. DHS official 1,930,475 (shortfall of 20,000 or 1.05%).
- **1914 row**: country columns sum to 1,059,391, but reported Europe total is 1,058,391 (excess of 1,000).
- Not all country-decade combinations have been verified; only Italy, Ireland, and Germany were checked.

Impact on results. The Italy error understates Italian immigration in the instrument's "shift" component for 1900–1909. Since Italy accounted for ∼23% of European immigration in that decade, and the settlement shares are fixed at 1900, the bias propagates to every city with Italian immigrants. The direction of bias depends on the correlation between Italian settlement shares and other factors. At minimum, this introduces noise; at worst, it could account for part of the significant negative reduced-form effect.

Recommendation: Re-digitize the Historical Statistics data with manual verification before treating these reduced-form results as confirmed.

6 Limitations

1. **Digitization error in immigration data.** The confirmed 20,000 undercount for Italy 1900–1909 and the 1914 discrepancy undermine confidence in the instrument. This is the most urgent issue.

2. **Weak first stage.** Annual flow instrument does not predict Census-measured immigrant stocks well (KP $F = 1.03$), making 2SLS unreliable. The reduced form remains valid as an intent-to-treat estimate.

3. **Sensitivity to controls.** All reduced-form effects vanish with baseline controls, raising concern about whether controls absorb confounding or absorb identifying variation.

4. **Annual instrument variation.** Within-decade annual variation may be partly mechanical (proportional to aggregate European flows) rather than reflecting genuine cross-country composition shifts. The instrument's identifying power at annual frequency may differ from the decadal case.

7 Human–AI Collaboration

Human researcher (Guohui Jiang):

- Designed the annual panel strategy (prompted by Round 1's power limitation)
- Specified the identification approach: annual shift-share from Historical Statistics × IPUMS shares
- Provided IPUMS account access and authorized extract submission
- Identified the Lyons hedonic indices as additional outcome measures

- Directed post-hoc data verification
- Interpreted the weak first stage and decided the reduced form is the primary result

Claude Code (AI assistant):

- Digitized the Historical Statistics annual immigration data via OCR **(with errors — see Section 5)**
- Adapted Lyons HPI/RPI code for city-level hedonic index construction (10 build scripts)
- Generated IPUMS API key, submitted and downloaded two complete-count extracts (∼11 GB total)
- Processed IPUMS 1900 data (∼76M records) for settlement shares
- Constructed the annual shift-share instrument
- Ran all regressions (reduced form, first stage, 2SLS, OLS)
- Verified digitized data against DHS decade totals (identified the Italy discrepancy)
- Drafted this progress report and research log entries

Key lesson from this round: The AI assistant can build complex empirical pipelines rapidly (13 scripts in one session), but **data digitization from scanned historical sources remains error-prone**. The Italy error was only caught through cross-validation against an independent source (DHS Yearbook). Automated OCR of historical statistical tables is not yet reliable enough to use without manual verification — especially when the digitized data enters the instrument, where errors propagate to all downstream estimates.

References

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